

Exploratory Data Analysis for Framingham Heart Disease Dataset

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1 Executive Summary

This report is an exploratory data analysis derived from the Framingham Heart Study, spanning across 4,240 cases. The dataset captures 14 variables across patient demographics, lifestyle factors, medical history, and clinical measurements. This analysis examines the distribution, relationships, and patterns to demonstrate the trends of the dataset. While this study includes some variables with hundreds of missing data, some preliminary findings can be drawn from it regarding the development of coronary heart disease (CHD).

In this report, exploratory findings show that this study had a population with the average age of 49.6 years old, and a slight bias toward a female majority (57.1%). The data shows a 10-year CHD prevalence of 15.2%, with those who are affected being notably older, and having a mean total cholesterol that falls in the borderline high range. Additionally, there is some moderate positive correlation between age and total cholesterol, as well as age and glucose level. We also see cigarettes smoked per day have a weak negative correlation with age a weak positive correlation with heart rate. When chi-square tests are performed, most categorical variables are significantly associated with CHD except for the number of cigarettes smoked per day.

2 Dataset Overview

2.1 Structure

The dataset contains 4240 unique rows of data across fourteen variables. All variables are stored as numeric values, with eight categorical variables (gender, education, current smoker, blood pressure medication, coronary heart disease, hypertension, diabetes, and stroke). The remaining six variables (age, cigarettes smoked

per day, total cholesterol, BMI, heart rate, and glucose) are continuous variables. Within this dataset there are numerous variables that have missing values in them, primarily glucose level. This indicates that some people in the sample may not have been tested for their glucose level, this will impact analysis of glucose as a contributing factor to Coronary Heart Disease (CHD). Glucose has the most missing values ($n = 388$), suggesting an inconsistency with how often participants are tested for glucose. Below is a summary of the dataset's variables including how many missing variables are in each field.

Table 1: Data summary

Name	mydata
Number of rows	4240
Number of columns	14
Column type frequency:	
factor	8
numeric	6
Group variables	None

Variable type: factor

skim_variable	n_missing	complete_rate	ordered	n_unique	top_counts
gender	0	1.00	FALSE	2	Fem: 2420, Mal: 1820
education	105	0.98	FALSE	4	Ele: 1720, Hig: 1253, Col: 689, Gra: 473
currentSmoker	0	1.00	FALSE	2	No: 2145, Yes: 2095
BPMeds	53	0.99	FALSE	2	No: 4063, Yes: 124
prevalentStroke	0	1.00	FALSE	2	No: 4215, Yes: 25
prevalentHyp	0	1.00	FALSE	2	No: 2923, Yes: 1317
diabetes	0	1.00	FALSE	2	No: 4131, Yes: 109
TenYearCHD	0	1.00	FALSE	2	No: 3596, Yes: 644

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
age	0	1.00	49.58	8.57	32.00	42.00	49.0	56.00	70.0	
cigsPerDay	29	0.99	9.01	11.92	0.00	0.00	0.0	20.00	70.0	
totChol	50	0.99	236.70	44.59	107.00	206.00	234.0	263.00	696.0	
BMI	19	1.00	25.80	4.08	15.54	23.07	25.4	28.04	56.8	
heartRate	1	1.00	75.88	12.03	44.00	68.00	75.0	83.00	143.0	
glucose	388	0.91	81.96	23.95	40.00	71.00	78.0	87.00	394.0	

2.2 Variable Summary

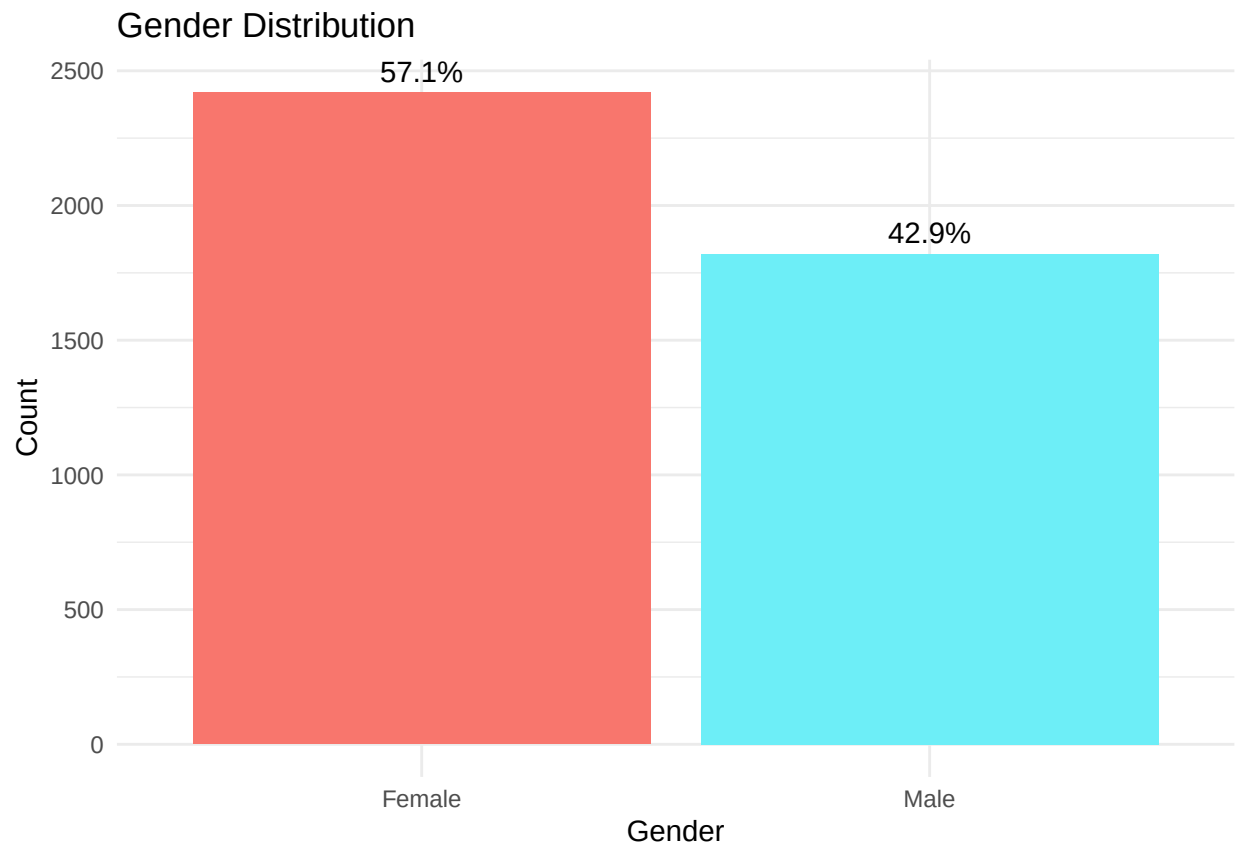
Table 2: Dataset Variable Summary

Variable	Type	Description	Missing
gender	Categorical	Gender of respondent: 0 = Female	None
age	Continuous	Age of respondent	None
education	Categorical	Education level	Yes
currentSmoker	Categorical	Current cigarette smoker	Yes
cigsPerDay	Continuous	Cigarettes smoked per day	Yes
BPMeds	Categorical	On blood pressure medication	Yes
prevalentStroke	Categorical	History of stroke	Yes
prevalentHyp	Categorical	History of hypertension	Yes
diabetes	Categorical	Diagnosed with diabetes	Yes
totChol	Continuous	Total cholesterol	Yes
BMI	Continuous	Body Mass Index	None
heartRate	Continuous	Heart Rate	Yes
glucose	Continuous	Glucose level	Yes
TenYearCHD	Categorical	10-year risk of CHD	Yes

3 Demographic Analysis

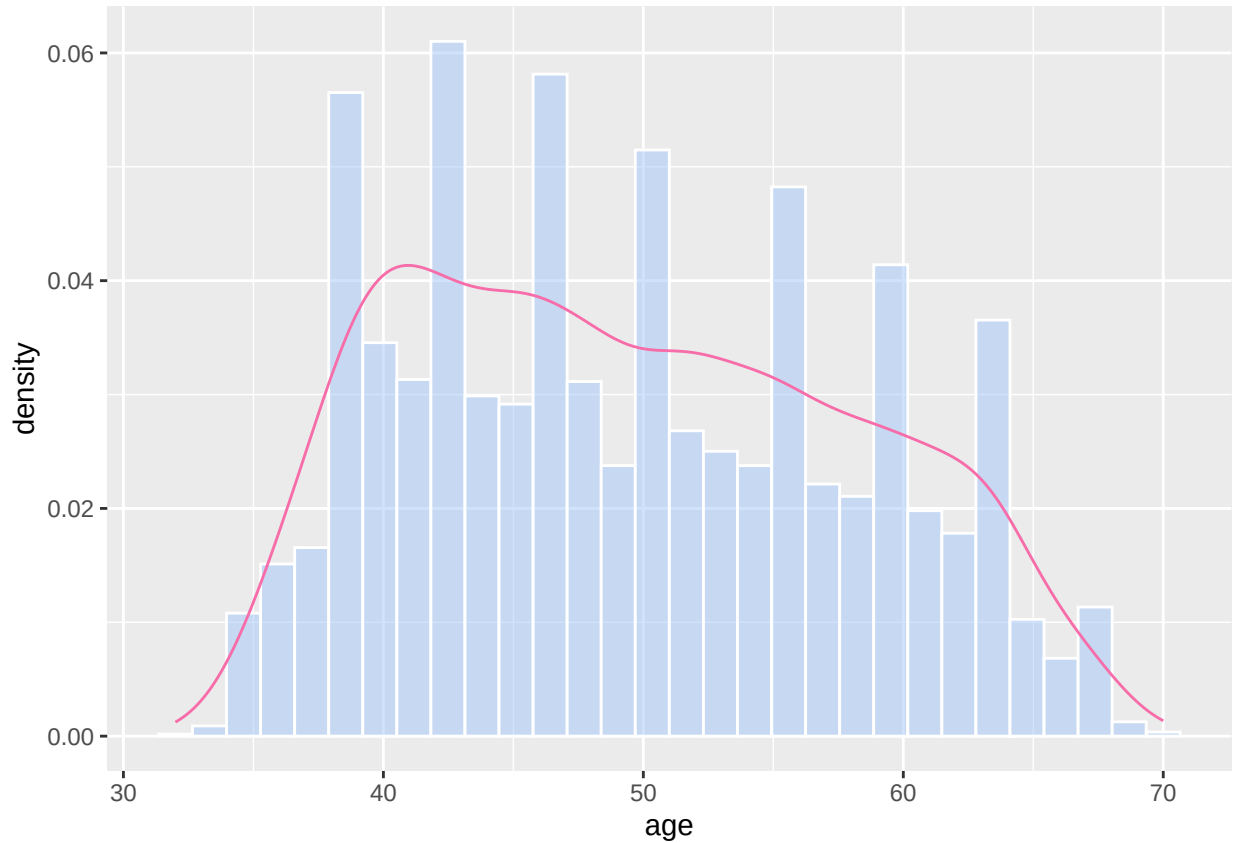
3.1 Gender Distribution

The dataset contains an overrepresentation of women ($n = 2,420$; 57.1%), and a less than representative sample of men ($n = 1,820$; 42.9%). This imbalance may skew analysis of associations between health status and gender. There are no missing values for the categorical variable **gender**.



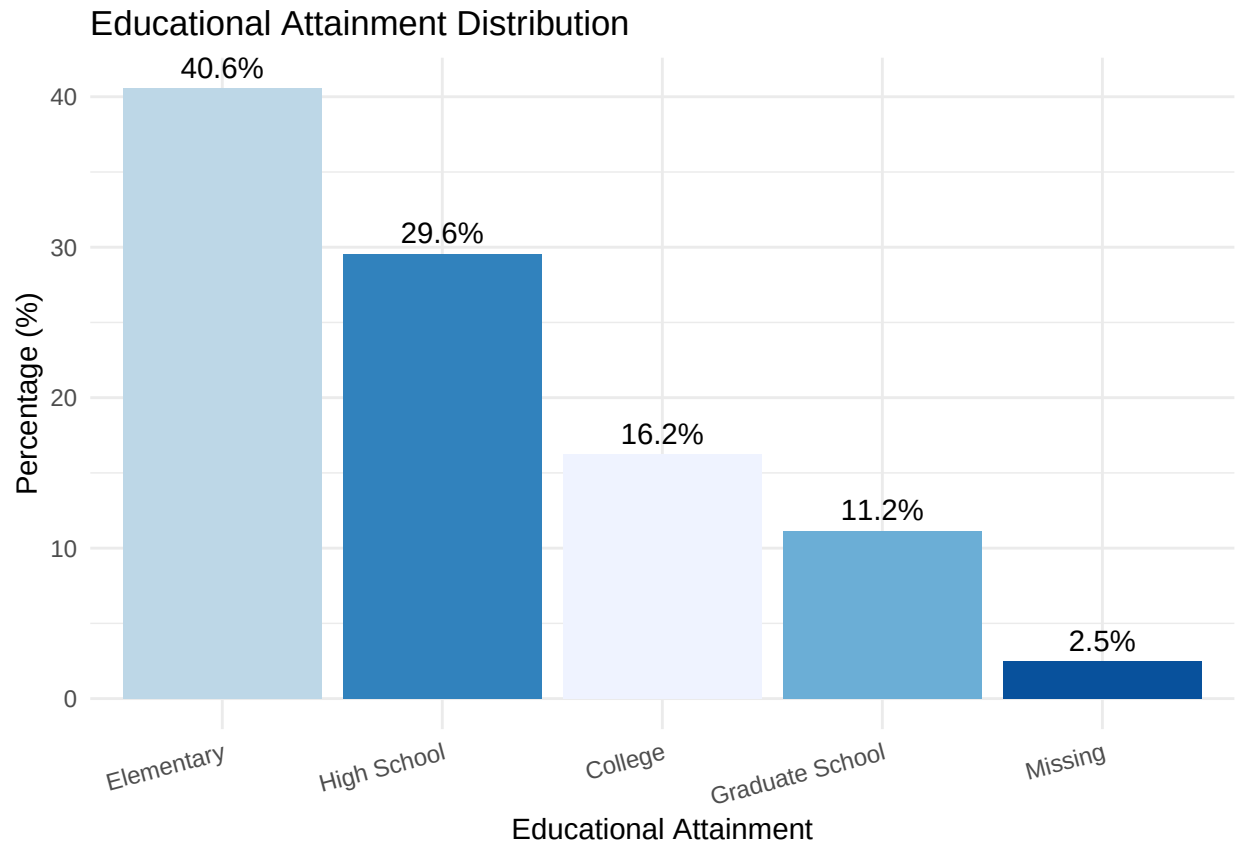
3.2 Age Distribution

Ages within this dataset range from 32 to 70 years old and has a mean age of 49.58. Additionally, age in the dataset has a standard deviation of 8.57, and a median age of 49. The older ages in this sample indicate that the study likely focused on older adults who are at risk for cardiovascular diseases. There are no missing values for the continuous variable `age`.



3.3 Educational Attainment

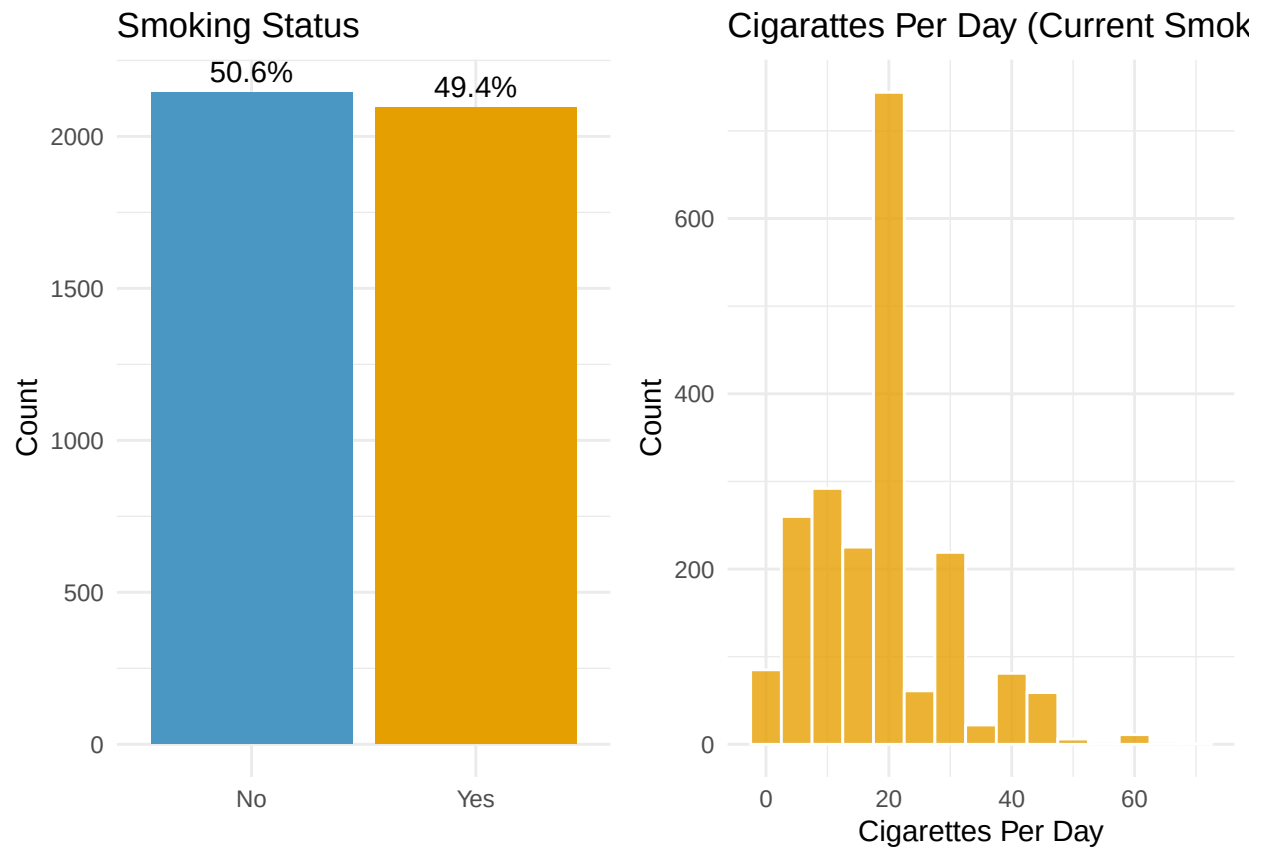
The cases in this dataset have a diversity of educational attainment with the largest group being less than high-school ($n = 1,720$; 40.6%), followed by high-school ($n = 1,253$; 29.6%), then college ($n = 689$; 16.2%), and finally graduate school ($n = 473$; 11.2%). A notable amount of people in the study did not report their educational attainment for this categorical variable ($n = 105$; 2.5%). This high amount of less than high-school educational attainment is not aligned with the percentage of people who have attained an education less than a high school graduate according to the US Census Bureau.



4 Health Risk Factor Analysis

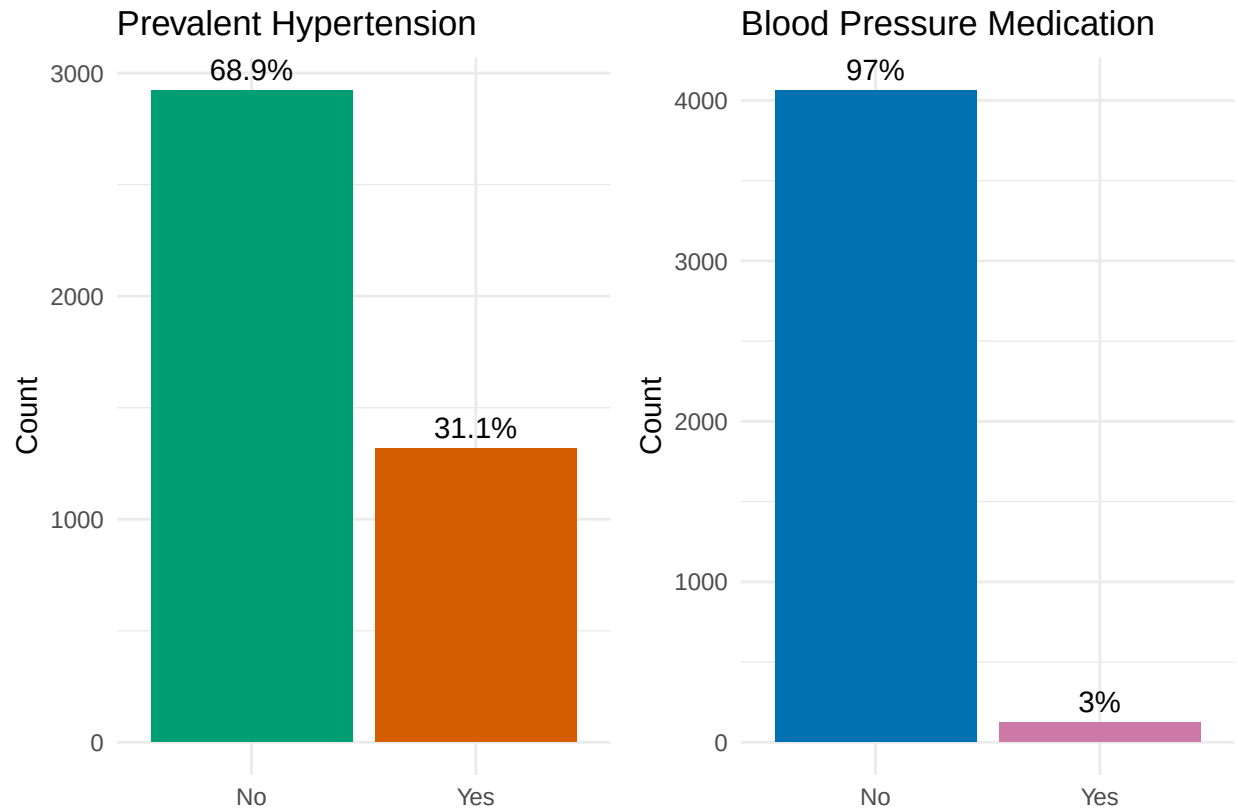
4.1 Smoking Behavior

The study is nearly evenly divided between smokers (49.4%) and non-smokers (50.6%). This indicates a high number of smokers included in the study, and among smokers, their consumption ranges from 1 to 70 cigarettes per day. On average, smokers smoke 18.36 cigarettes daily and a median of 20 cigarettes and standard deviation of 11.92.



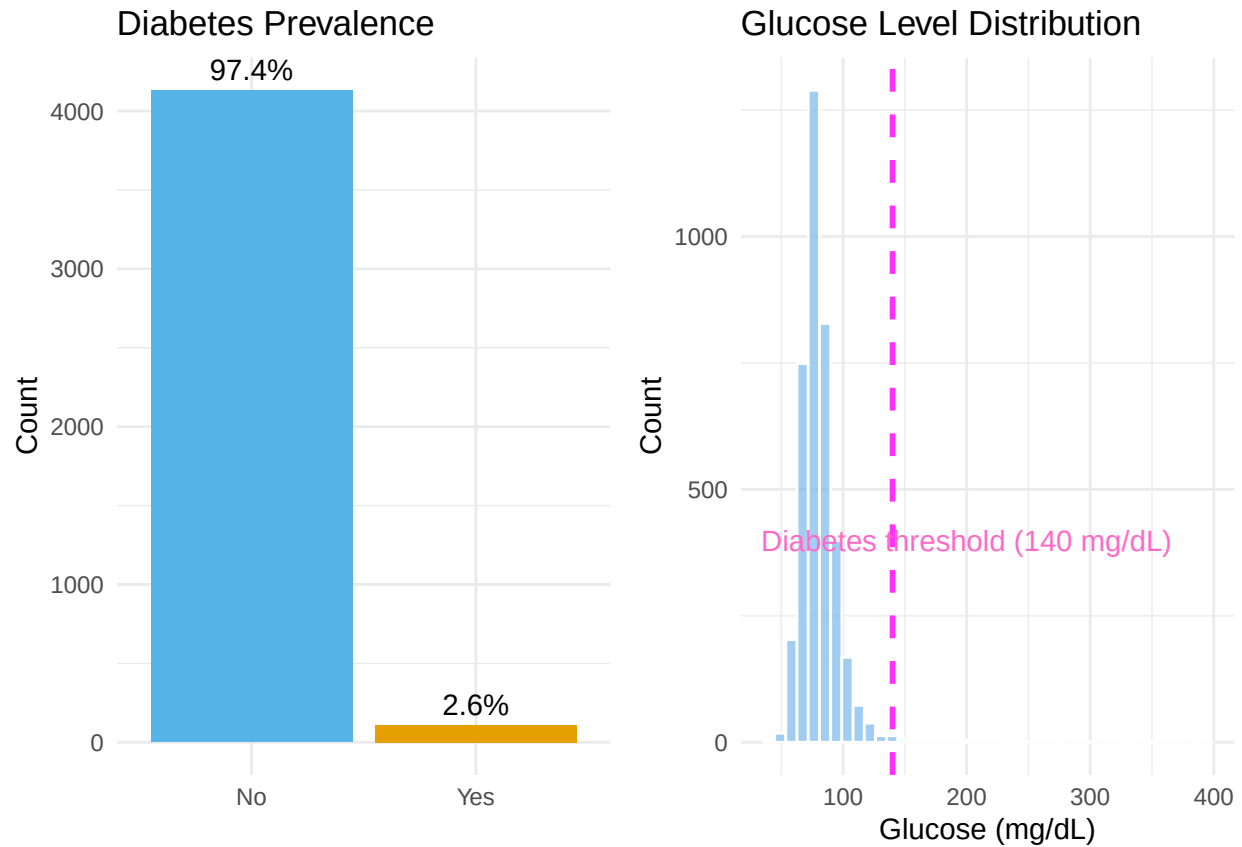
4.2 Blood Pressure and Hypertension

Hypertension is prevalent in 31.1% of the cases in this dataset, while only 3% of the dataset are on blood pressure medication. This disparity suggests that the population in the study is has either untreated hypertension or indicates that hypertension variables captures disgnosed hypertension regardless of medication status. There may be potential for missed reduction in cardiovascular disease.



4.3 Diabetes and Glucose Levels

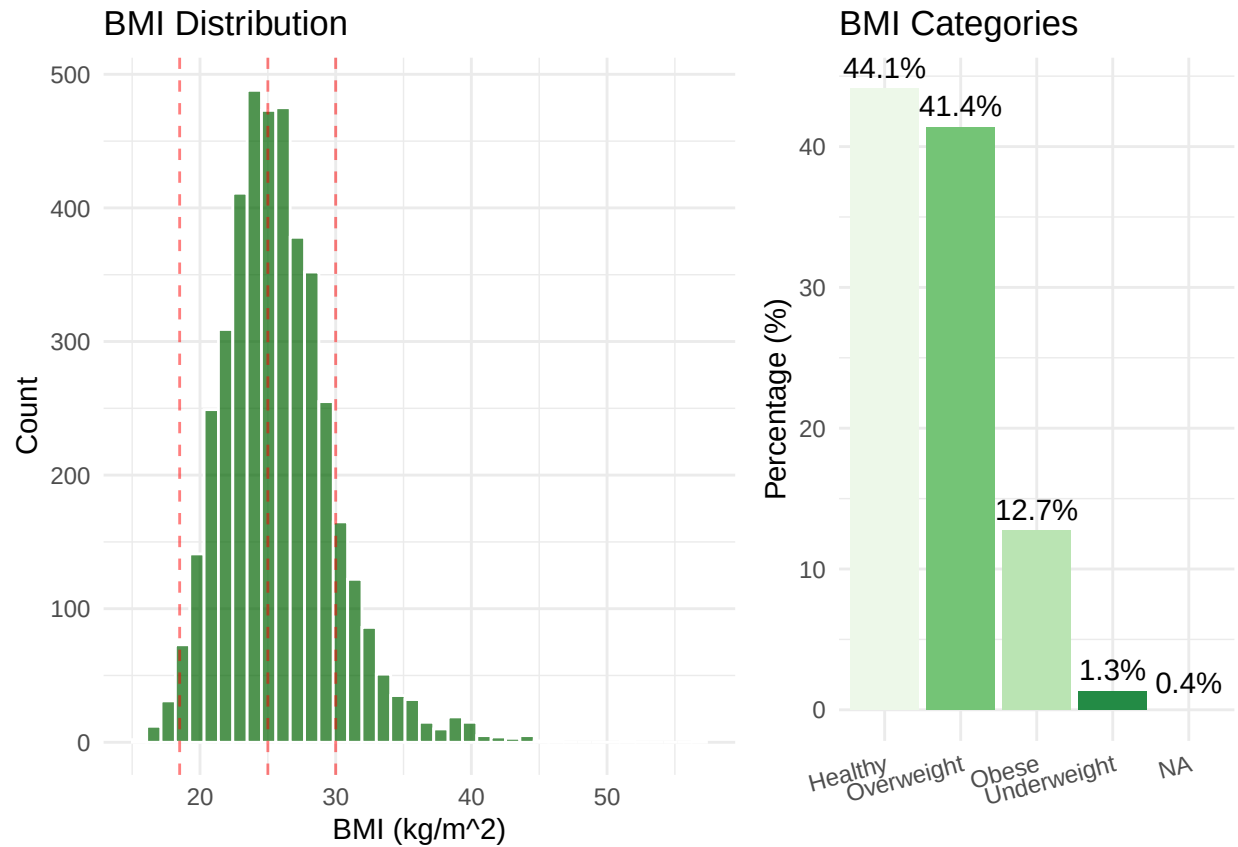
2.6% of the study's population has diabetes which is consistent with the data from the glucose variable. The distribution glucose levels is shows a pronounced right skew due to diabetic patients having elevated glucose readings. Within this exploratory data analysis, the threshold for using glucose levels to determine diabetes (140 mg/dL) is based on the Mayo Clinic's reporting. The variable `glucose` has 388 missing values, which is a high percentage of the dataset (9.2%). It has mean of 81.96, standard deviation of 23.95, and ranges from 40 to 394 mg/dL.



5 Clinical Measurements

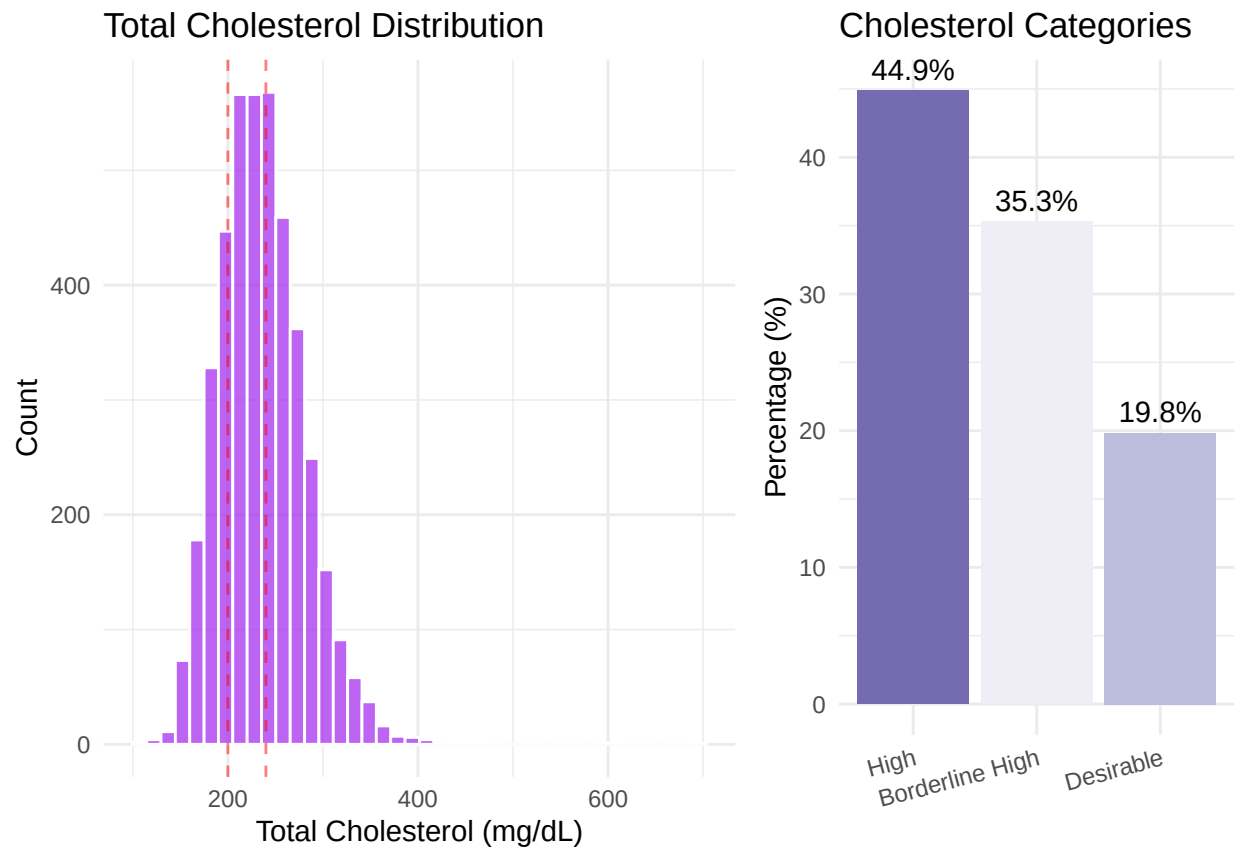
5.1 Body Mass Index (BMI)

BMI values (in kg/m^2) range from 15.54 to 28.04 with a mean of 25.80 and a standard deviation of 4.08. This high mean and below bar chart show that the majority of people in this study have a high BMI that can be associated with increased risk of heart disease. BMI categories are based on the CDC's guidelines.



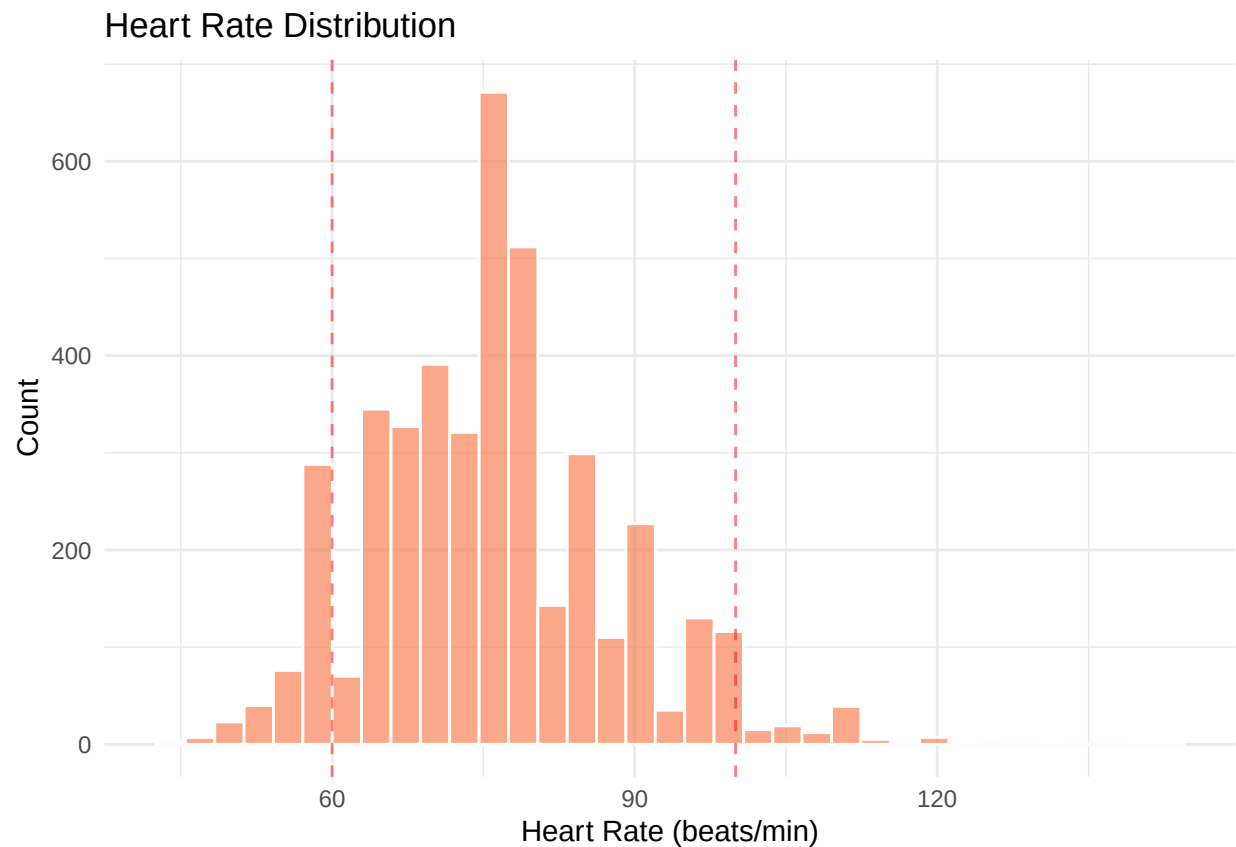
5.2 Cholesterol

A high cholesterol level according to the Mayo Clinic is around 240 mg/dL with below 200 mg/dL being desirable. The distribution within the study shows that nearly half had high levels of cholesterol. This may be indicative of the higher risk of heart disease associated with higher cholesterol.



5.3 Heart Rate

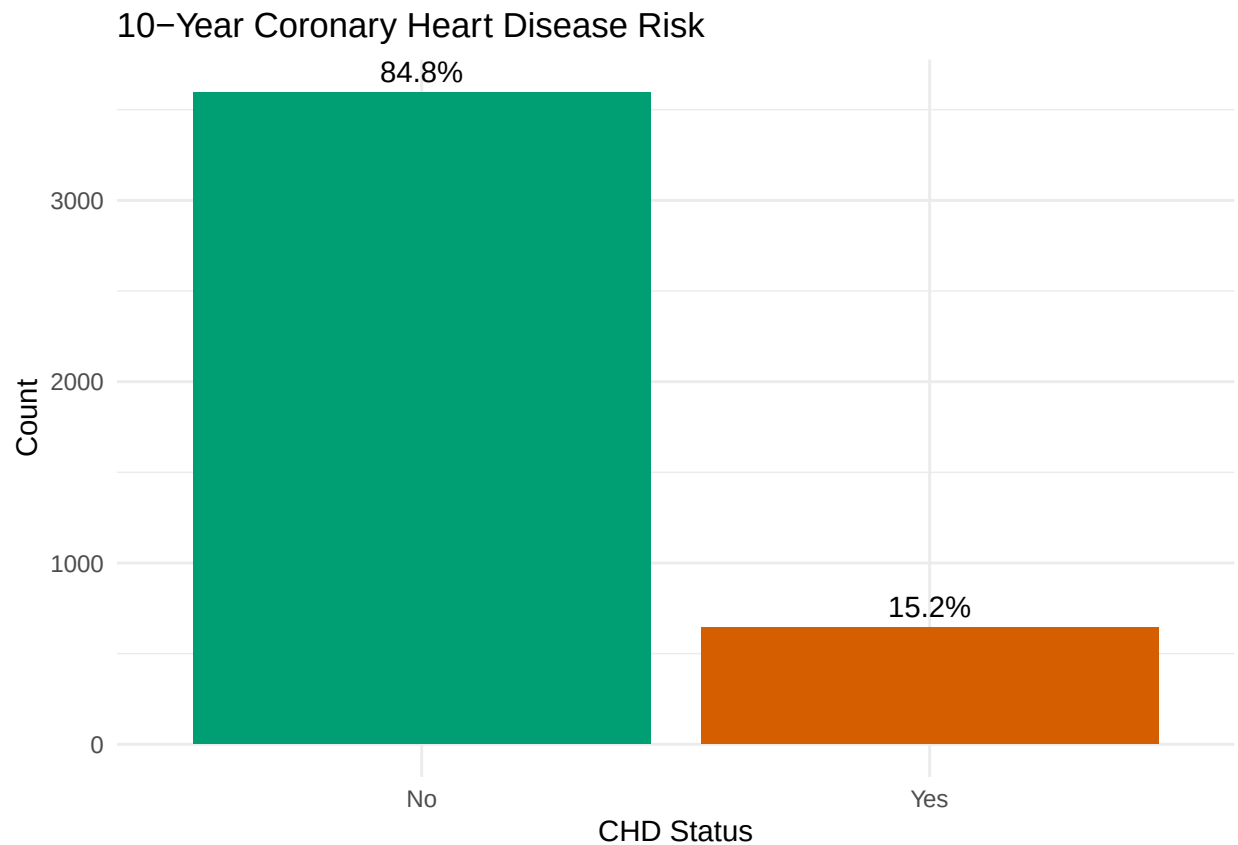
The resting heartrate ranges from 44 to 143 beats per minute, with a mean of 75.9, and a median of 75, and standard deviation of 12.03. The distribution is somewhat normal with a slight right skew. The mayo clinic reports that a normal resting heart rate is between 60 and 100 beats per minute. Some participants in the study have high resting heart rates, which may be indicative of higher risk of heart disease. With low missing data and a normal distribution, using the variable `heartRate` in regression models would prove useful.



6 Coronary Heart Disease (CHD) Analysis

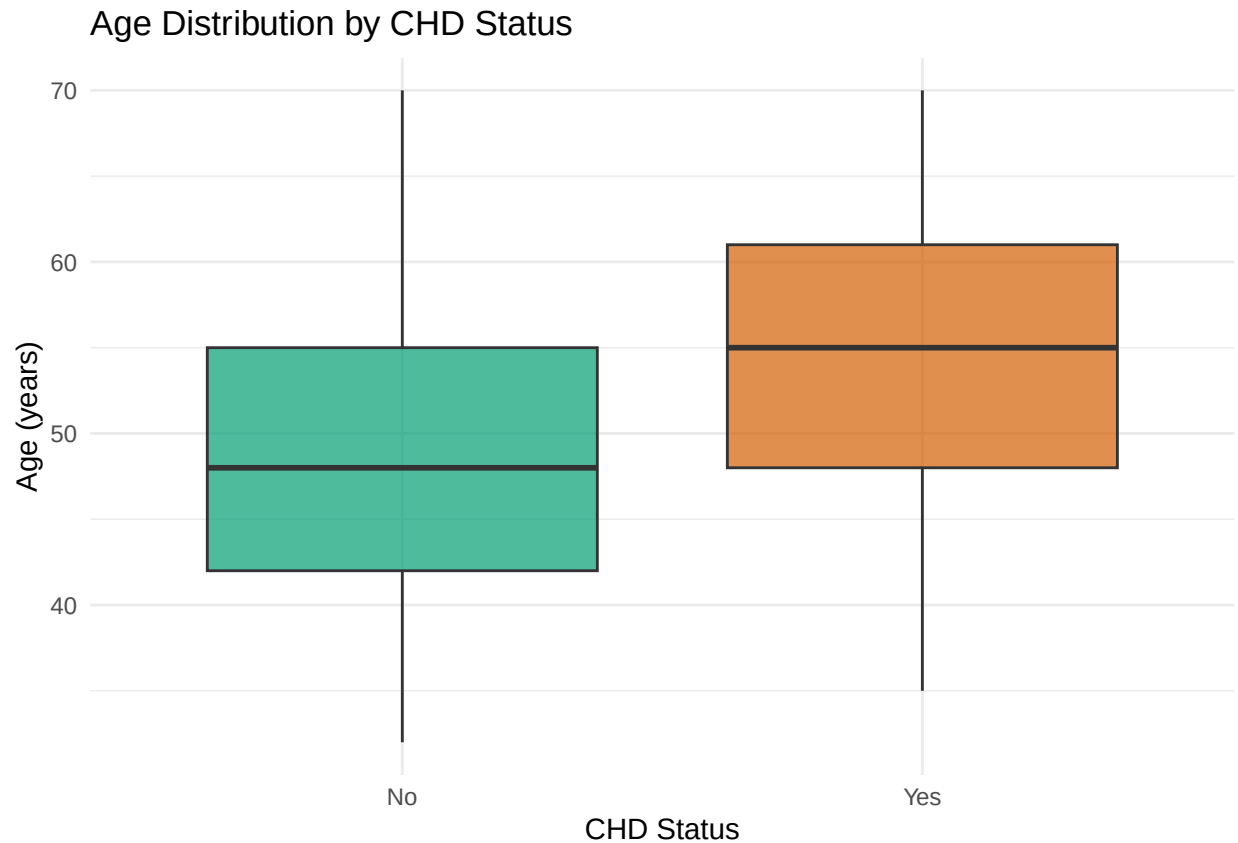
6.1 CHD Prevalence

Coronary heart disease is a categorical variable with no missing values in the dataset. Within the study, 3,596 (84.4%) people did not have CHD, while 644 (15.2%) people do. The variable `TenYearCHD` is a binary variable with no missing values.



6.2 Age and CHD Risk

When looking at age distribution by CHD status, we can see that the mean of those without CHD is lower than those with CHD. This tracks well with increased heart disease risks as people age. The mean age of people without CHD is 48.8 ($n = 3596$; $sd = 8.41$), and the mean age for those with CHD is 54.1 ($n = 644$; $sd = 8.01$). T-tests may be useful to see if there is a statistically significant difference in the means of each group. There are no missing values both `TenYearCHD` and `age`.



6.3 Risk Factors by CHD Status

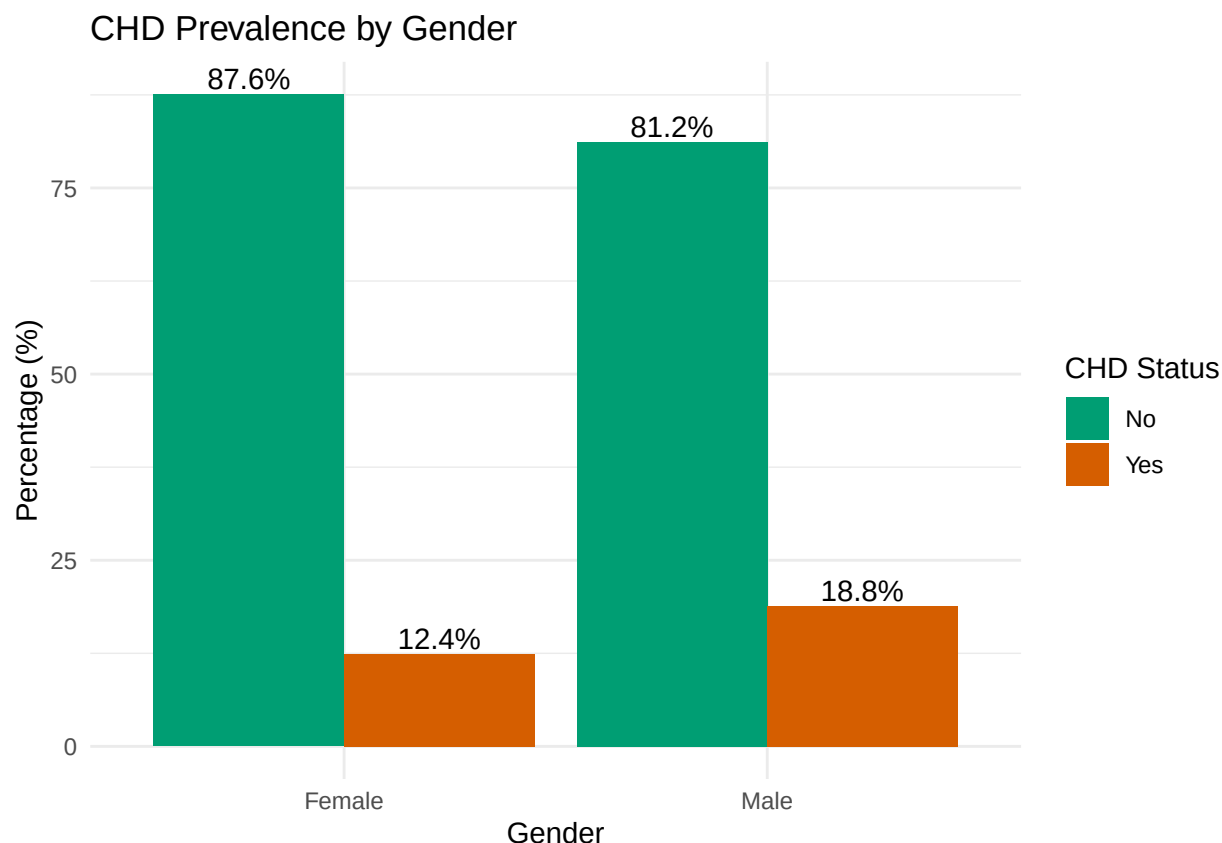
When looking at common risk factors for CHD, we see that for those without CHD, the mean of each risk factor is lower across the board. These results are displayed in Table 3: Risk Factors by CHD Status.

Table 3: Risk Factors by CHD Status

Variable	CHD_No	CHD_Yes
Count	3596.0	644.0
Age	48.8	54.1
Cigs/Day	8.7	10.6
Glucose	80.7	89.0
BMI	25.7	26.5
Cholesterol	235.1	245.4
Heart Rate	75.8	76.5
Hypertension (%)	27.6	50.5
Diabetes (%)	1.9	6.2
Smoker (%)	49.0	51.7

6.4 Gender and CHD

Keeping in mind the overrepresentation of women in the dataset, we see that CHD prevalence is lower for women than it is for men. This is consistent with the what the National Institute of Health reports for the ages in this dataset. However, has Harvard Health Publishing states there is a significant gap in women and men for identifying and treating heart disease. There are no missing values for the categorical variables `gender` and `TenYearCHD` which lends itself well to further analysis.

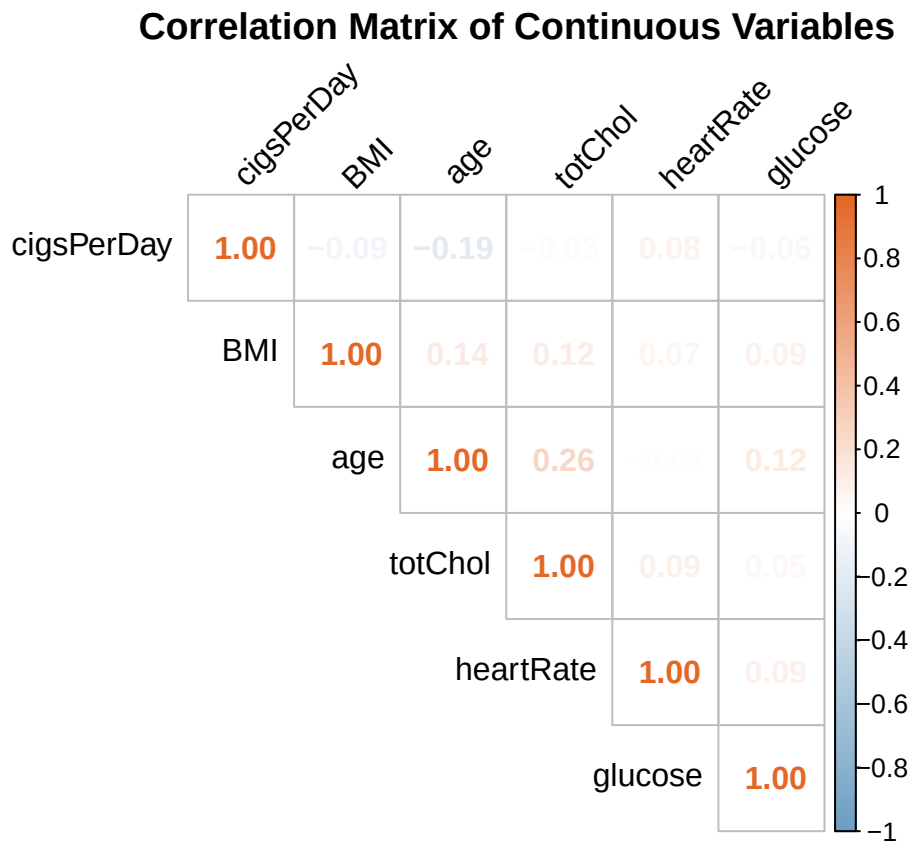


7 Correlation Analysis

7.1 Correlation Matrix of Continous Variables

In a correlation matrix of the continous variables (`cigs/day`, `BMI`, `age`, `total cholesterol`, `heart rate`, and `glucose`), we see that there is a moderate positive correlation between `age` and `total cholesterol` and `age` with `glucose level`. These relationships, however, remain fairly week. A negative relationship exists between `cigarettes smoked per day` and `age`, indicating that older people in this study smoke less than younger

participants. Additionally, there is a extremely slight negative correlation between cigarettes smoked per day and BMI.

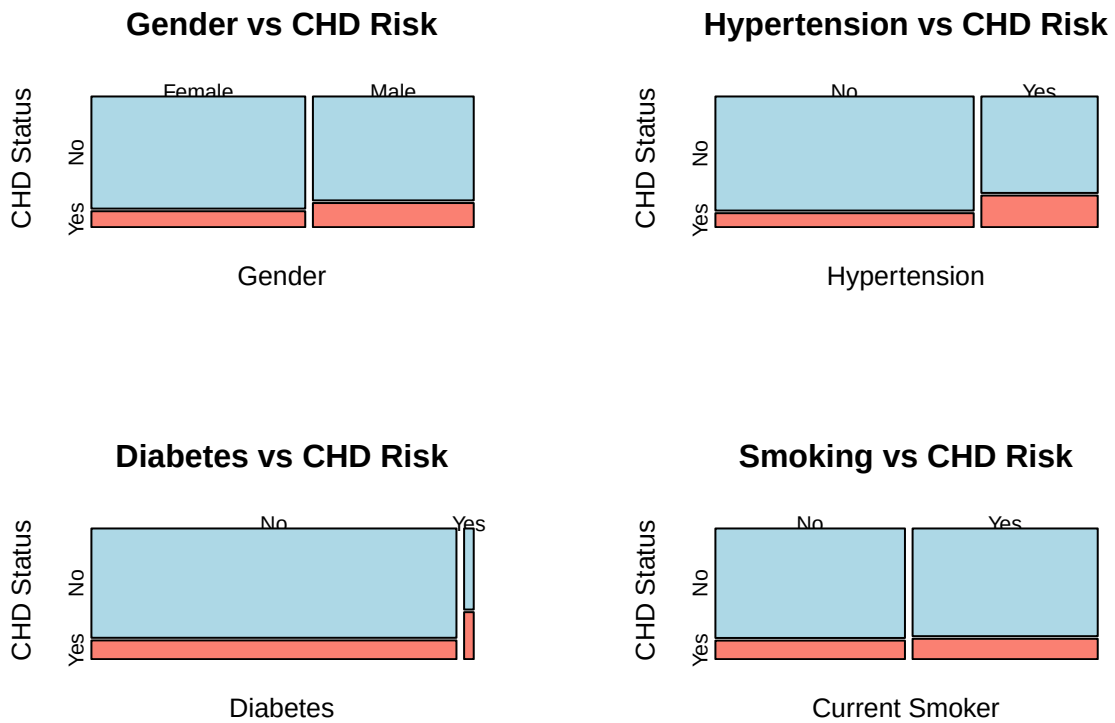


7.2 Chi-Square Test for Categorical Variables

As this dataset contains multiple categorical variables, a chi-square test can illuminate some relationships between them. In Table 4 below, when operationalizing 10-year CHD risk as the dependent variable, most categorical variables (**gender**, **education**, **blood pressure meds**, **prevalentStroke**, **prevalentHyp**, and **diabetes**) ($p\text{-values} < 0.05$) all have a statistically significant relationship positive relationship with a participant having CHD. The only categorical variable that does not have a significant association with CHD is **currentSmoker** ($p = 0.22$). Further analysis using regressions might illuminate a more complex relationship between variables and their impact on CHD. Some of these chi-square tests are visualized below.

Table 4: Chi-Square Tests: Categorical Variables vs 10-Year CHD Risk

	Variable	Chi_Square	df	p_value	significant
X-squared	gender	32.62	1	1.122e-08	TRUE
X-squared1	education	32.02	3	5.19e-07	TRUE
X-squared2	currentSmoker	1.50	1	0.2211	FALSE
X-squared3	BPMeds	30.65	1	3.097e-08	TRUE
X-squared4	prevalentStroke	14.03	1	0.0001796	TRUE
X-squared5	prevalentHyp	132.46	1	< 2.2e-16	TRUE
X-squared6	diabetes	38.48	1	5.525e-10	TRUE

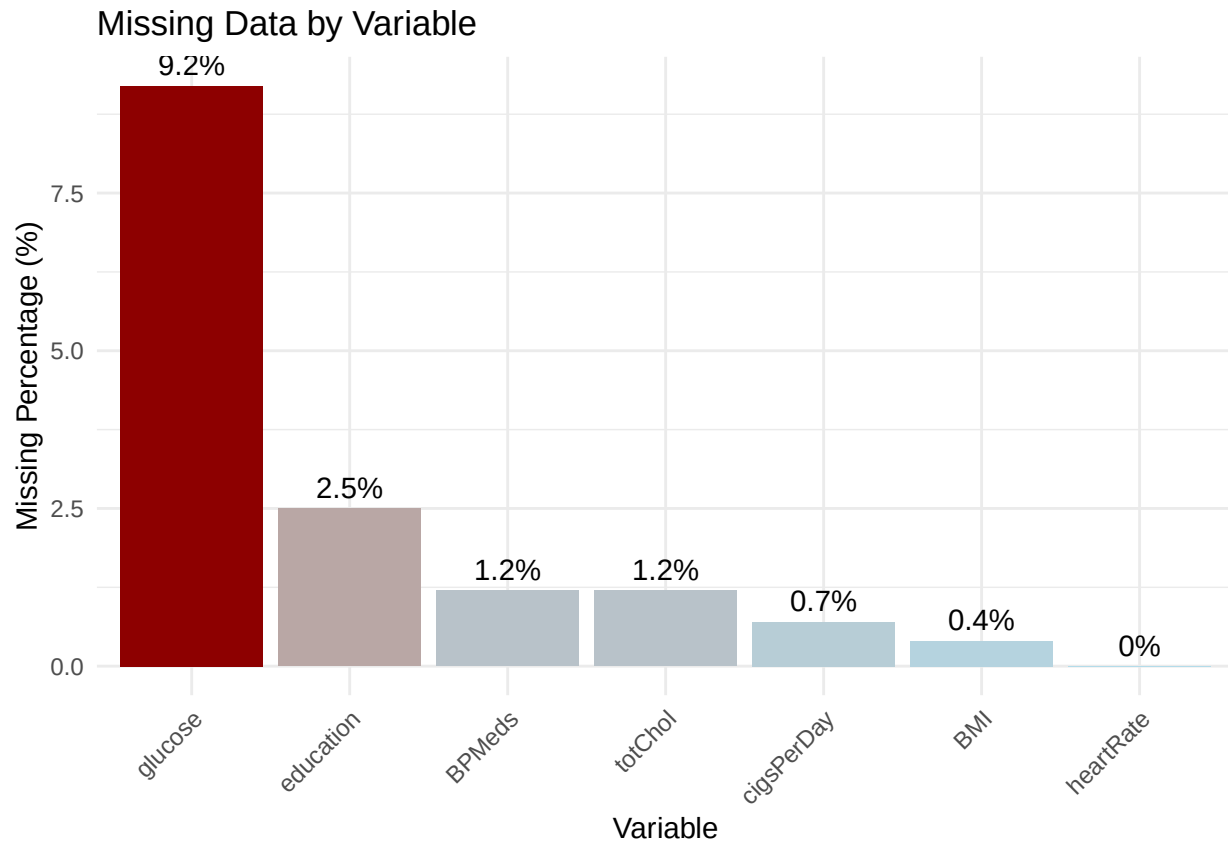


8 Missing Data

8.1 Missing Value Patterns

The below chart demonstrates each variable that has missing data and their percentages. The variable `glucose`, as previously noted, has the most missing data ($n = 388$; 9.2%) within the dataset. This may be due to collection within the study and some patients not needing their glucose tested, but may also

be a lapse in identifying possible CHD within participant. The variable **education** has the second most missing data ($n = 105$; 2.5%), which may be due to collection errors or participants not willing to report their educational attainment. Further analysis might consider treating these missing data as an unknown category, or dropping these cases from larger regression models.



9 Conclusion

This exploratory data analysis of the Framingham Heart Study dataset yields some important preliminary insights into factors associated with CHD. First, this analysis confirms the established risk factors associated with CHD. Especially with age showing consistent differences between CHD and non-CHD groups (5.3 year difference in mean ages). Hypertension also shows a strong relationship with CHD ($X^2 = 132.46$; $p\text{-value} < 0.001$). And while diabetes was relatively rare in the overall population of the study (2.6%), participants with CHD are significantly likely to have it ($X^2 = 38.48$; $p\text{-value} < 0.001$). This association highlights the potential risk of CHD for people with diabetes.

The female majority population alongside the lower CHD prevalence for women (12.4% vs. 18.8% in men)

aligns with the established gap in identifying heart disease in women, and the higher rate of CHD in men for this age cohort. Educational attainment in this dataset shows a predominance of lower educational levels (40.6% elementary education) which is not consistent with the overall country. This may be reflect characteristics of the study's cohort, or potential socio-economic influences on increased cardiovascular risk.

The correlation matrix revealed modest correlations among continuous variables. these suggest that factors contribute relatively independently to CHD risk, which lends support to their inclusion in more robust models. An interesting relationship is the negative correlation between age and cigarettes smoked per day, which might reflect the stopping of smoking with age or age cohort effects on smoking.

Overall, this dataset has some concerning gaps in recorded data, particularly glucose, which would be vital information to assess diabetes, a known risk factor for CHD. This is a significant limitation for analysis, may potentially introduce bias into statistical testing. While this variable is low, the complete data on 10-year CHD is a strength for predictive modeling, and most clinical measurements have low missing rates (generally below 2%). The dataset is has some considerable strength in this regard, but better data collection for glucose levels would lead to more complete analyses.

For future analyses, this report lends support to the need for multivariate risk prediction models that span variables of gender, age, hypertension, diabetes, cholesterol, and BMI. The non-significant chi-square test results for current smoking ($p = 0.22$) may suggest that other habits around smoking should be considered in the data collection that could reveal more granular insights than merely current smoking status.

In all, this exploratory analysis provides an introduction and characterization to the Framingham Heart Study dataset. Many of the tables and plots contained within reveal patterns consistent with established literature on CHD. This report may be useful in subsequent predictive modeling for risk factor analysis, and despite some missing data, the dataset has some strength with many variables containing complete data, many risk factors recorded, and a large number of participants. This dataset is ripe for further analysis, as demonstrated in this report.